

AREF BAK ALI

AI Engineer | Agentic AI | LLMs, RAG, Generative AI | MLOps

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Summary

Second-year Master's student (M2) in Artificial Intelligence, Data Science and Agentic AI at Université Paris Dauphine-PSL, driven by projects where AI becomes a real product: useful, reliable and well integrated. My experience covers AI agents, RAG, Large Language Models (LLMs), NLP, Machine Learning and MLOps. I enjoy learning, experimenting and understanding what actually works in practice. Seeking a 6-month end-of-studies internship starting February 2027 in AI Engineering, Machine Learning, Data Science or Agentic AI.

Skills

Generative & Agentic AI: LangGraph (agents, tool calling), RAG, LLM serving, Hugging Face Transformers, Vector DB (Qdrant).
ML & Deep Learning: PyTorch, TensorFlow/Keras, Scikit-learn, XGBoost, CamemBERT, Computer Vision (OpenCV, MediaPipe).
Programming & Data: Python (FastAPI, Pandas, NumPy), R, SQL (MySQL, SQL Server).
MLOps & Tools: Docker, CI/CD (GitHub Actions), MLflow, Evidently, Git.

Education

Université Paris Dauphine-PSL – Tunis Campus **09/2025 – 2027 (expected)**
Master's in Artificial Intelligence, Data Science and Agentic AI (M1 completed, M2 in progress)
Key courses: Machine Learning, Deep Learning, NLP & LLMs, Agentic AI, MLOps, Data Mining, Probabilistic Models.

Université Paris Dauphine-PSL – Tunis Campus **09/2022 – 05/2025**
Bachelor's in Mathematics Applied to Computer Science
Key courses: Applied Mathematics, Statistical Modeling, Probability Theory, SQL.

Professional Experience

AI Intern – Conversational Calendar Agent **07/2026 – 09/2026**
ARSII – Remote internship
- Designed a multi-tool AI agent (LangGraph), traced and evaluated with LangSmith: **87% accuracy** on 60 test cases.
- Experimented with vLLM for serving an open-source LLM, then used Ollama for local inference with **1.6 s average latency**, within an agent connected to the user's calendar and email.
  LangGraph, LangSmith, Tool calling, vLLM, Hugging Face, Ollama, FastAPI, React, TypeScript, Google Calendar API, Gmail API

Intern – Educational AI Agent (Math+ Program) **07/2026 – 09/2026**
Université Paris Dauphine-PSL, Tunis Campus
- Developed AI and autonomous agent features (**Gemini**, **Google Cloud/Vertex AI**) for a mathematics student support platform; integrated a **RAG** system to improve answer relevance and personalization.

Mobile Development Intern – Grow Up Technologies: *Comptazone* app (React Native). **06/2024 – 07/2024**
Web Development Intern – STE COMPT-INFO SERVICES: website for an accounting firm (PHP). **06/2023 – 07/2023**

Projects

BreastFriend – AI Oncology Assistant *(Generative AI / RAG)* **12/2024 – 10/2026**
- Built in 12/2024 and revisited in 10/2026, this medical RAG chatbot won the **Business Impact Award** (*Hack for Good with Gen AI*).
- Developed a hybrid RAG pipeline with reranking, evaluated with Ragas.
- Integrated emergency detection, patient-doctor follow-up and virtual try-on (computer vision).
  Python, Flask, React, Qdrant, Ollama, Ragas, MediaPipe, OpenCV, Docker

PostureAI – Real-Time Posture Assistant *(Computer Vision)* **09/2026**
- Local posture-correction assistant using computer vision (**MediaPipe**): head/neck/shoulder tracking, personalized calibration, smart alerts and progress dashboard; 100 % on-device processing.
  Python, OpenCV, MediaPipe

AI Platform for Breast Cancer Prediction *(Machine Learning & MLOps)* **05/2026**
- End-to-end MLOps pipeline: experiment tracking (**MLflow**), data drift detection (**Evidently**), **Docker** containerization and **CI/CD**; compared 6 ML models (KNN selected, F1-score 81.00 %).
  Python, Scikit-learn, Streamlit, MLflow, Evidently, Docker, GitHub Actions

Drought Risk Forecasting in Tunisia *(Deep Learning)* **05/2026**
- ML/DL pipeline on Tunisian climate data (2017–2025): Random Forest ($\mathbf{R^2 = 0.821}$) then tuned LSTM ($\mathbf{R^2 = 0.953}$); risk mapping by governorate.
  Python, XGBoost, TensorFlow, Keras, GeoPandas

Credit Card Fraud Detection *(Data Mining)* **03/2026**
- Highly imbalanced dataset (fraud < 0.2 %): SMOTE resampling, supervised classification (**AUC-ROC = 0.98**), clustering and interpretability with SHAP/LIME.
  Python, Scikit-learn, SMOTE, SHAP

Activities & Awards

- **Vice-President, ITLAB Club** (AI Coordinator 2023–25).
- Organizing committee member for the *Hack for Good with Gen AI* and *Capture The Flag* hackathons.
- **First Prize, E5t Fund 2024** (environmental association) – *Ocean Wash* project: marine litter collection robot.
- **Third Innovation Prize**, Spring University 2023.
- Former international judoka – First place, Tunisian Championship.

Languages

Arabic: Native | French: Advanced (C1 – TCF) | English: Advanced (C1) | German: Elementary (A2)